



Operating System and System Programming (25CS2104E)

Linux File Descriptor and File I/O Management System

Review-I

Under the Esteemed Guidance of

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Abstract

- Our project is a Linux-based command-line application developed using C to demonstrate file descriptors and low-level file I/O. The system allows users to perform basic file operations such as creating/opening, reading, writing, appending, moving the file pointer, and closing files.
- Linux system calls such as `open()`, `read()`, `write()`, `lseek()`, and `close()` are used to perform these operations.
- The project demonstrates how a user application communicates with the Linux kernel and file system, and provides practical understanding of Operating Systems and Systems Programming concepts.

Introduction

File I/O refers to reading data from and writing data to files.

In Linux, an opened file is represented by a file descriptor, which is a non-negative integer assigned by the operating system kernel. The file descriptor is used by a process to perform operations such as reading, writing, seeking, and closing a file.

Standard File Descriptors

File Descriptor	Stream
0	Standard Input
1	Standard Output
2	Standard Error

Our project provides a simple terminal-based interface to demonstrate file operations and understand how applications interact with the Linux kernel through system calls.

Hardware and Software requirements

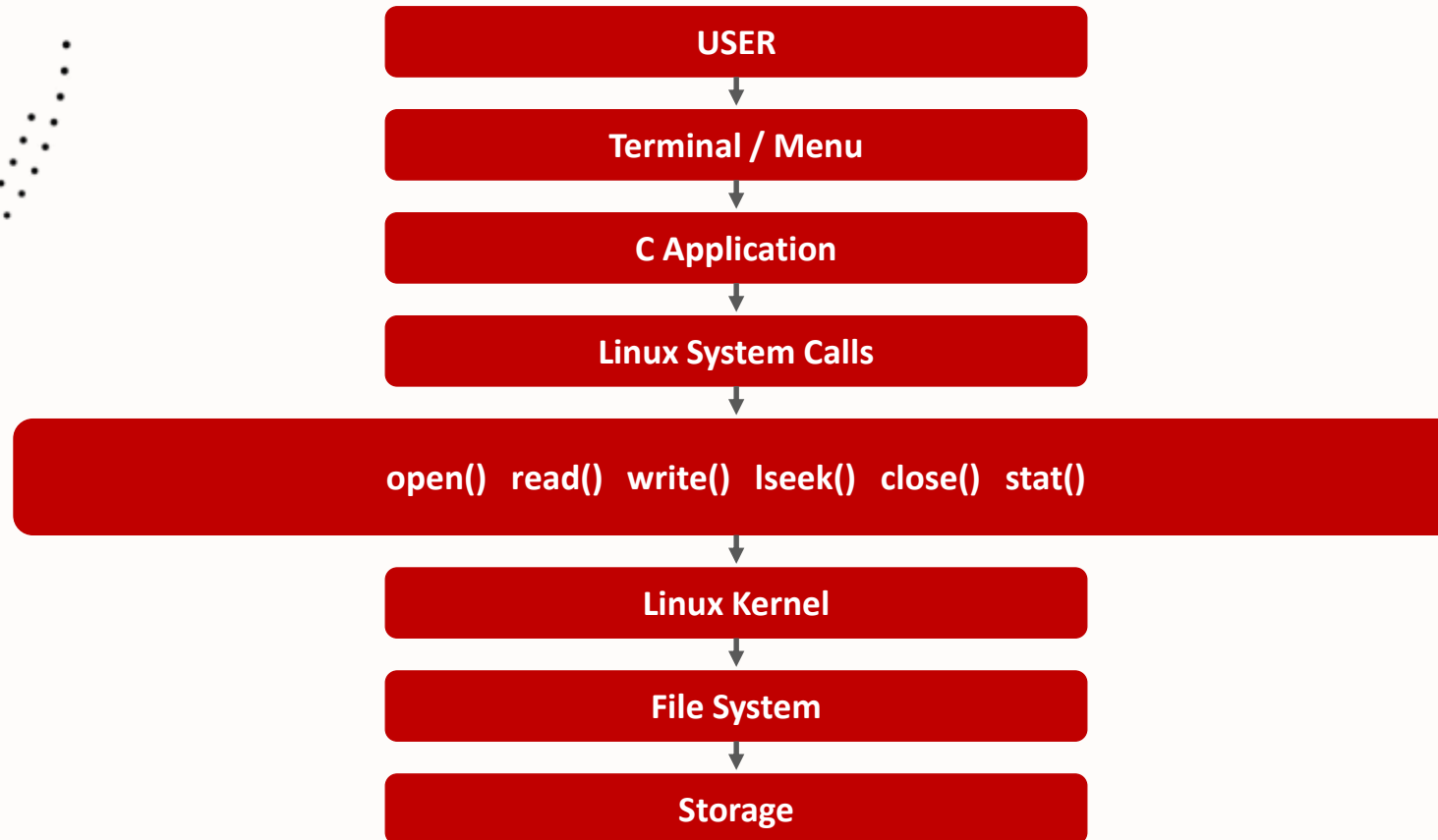
Hardware Requirements

Requirement	Specification
Computer	Laptop / Desktop
RAM	Minimum 4 GB
Processor	Minimum 1 GHz
Storage	Sufficient space for Linux and project files

Software Requirements

Software	Purpose
Ubuntu / Linux	Operating system and execution environment
C	Programming language
GCC	Compilation
Make / Makefile	Project build
VS Code / Vim	Code development
Git & GitHub	Version control and collaboration

System Architecture



Working Flow:

User Input → C Application → System Call → Linux Kernel → File System → Storage

The application sends file-operation requests through Linux system calls. The kernel manages the file descriptor and performs the requested operation on the file system.

Conclusion

The project demonstrates low-level file management in Linux using C. It provides practical understanding of file descriptors, system calls, file I/O, file offsets, permissions, and error handling. The system shows how a user-space application communicates with the Linux kernel to perform file operations.

Overall, the project connects theoretical Operating Systems concepts with a working Linux systems-programming implementation.

Thank you

Key Takeaway

File Descriptor → System Call → Linux Kernel → File System → Storage